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Sole structure for article of footwear

Abstract

A sole structure for an article of footwear includes a first annular group of traction elements arranged along a first annular zone and a second annular group of traction elements arranged along a second annular zone concentric with the first annular group. The first and second annular groups of traction elements include a plurality of directional traction elements arranged in a first rotational direction about a common rotation zone. Optionally, the first annular group of traction elements may include an omnidirectional traction element arranged at a location associated with a relatively low degree of alignment between radii of rotation corresponding to different torsional movements of the sole structure during use. The directional traction elements may include unidirectional traction elements or bidirectional traction elements at locations associated with moderate to high degrees of alignment between radii of rotation corresponding to the different torsional movements.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/470,790, filed on Sep. 9, 2021, which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 63/077,208, filed on Sep. 11, 2020. The disclosures of these prior applications are considered part of the disclosure of this application and are hereby incorporated by reference in their entirety.

FIELD

(1) The present disclosure relates generally to an article of footwear, and more particularly to a sole structure for an article of footwear.

BACKGROUND

(2) This section provides background information related to the present disclosure and is not necessarily prior art.

(3) Articles of footwear conventionally include an upper and a sole structure. The upper may be formed from any suitable material(s) to receive, secure, and support a foot on the sole structure. The upper may cooperate with laces, straps, or other fasteners to adjust the fit of the upper around the foot. A bottom portion of the upper, proximate to a bottom surface of the foot, attaches to the sole structure.

(4) Sole structures generally include a layered arrangement extending between a ground surface and the upper. For example, a sole structure may include a midsole and an outsole. The midsole is generally disposed between the outsole and the upper and provides cushioning for the foot. The

midsole may include a pressurized fluid-filled chamber that compresses resiliently under an applied load to cushion the foot by attenuating ground-reaction forces. The outsole provides abrasion-resistance and traction with the ground surface and may be formed from rubber or other materials that impart durability and wear-resistance, as well as enhancing traction with the ground surface. (5) While known outsoles have proven acceptable for their intended purposes, a continuous need for improvement in the relevant art remains. For example, a need exists for an outsole that provides improved traction with the ground surface when forces having varying magnitude and direction are applied from the midsole or the upper to the outsole. A need also exists for an article of footwear having improved overall comfort and fit while providing such improved traction.

Description

DRAWINGS

- (1) The drawings described herein are for illustrative purposes only of selected configurations and not all possible implementations, and are not intended to limit the scope of the present disclosure.
- (2) FIG. 1A is an environmental view illustrating a contact zone and centroid associated with an article of footwear and resulting from a first movement;
- (3) FIG. 1B is an environmental view illustrating a contact zone and centroid associated with an article of footwear and resulting from a second movement;
- (4) FIG. 1C is an environmental view illustrating a contact zone and centroid associated with an article of footwear and resulting from a third movement;
- (5) FIG. 1D is an environmental view showing the centroids associated with the first, second, and third movements overlaid onto an article of footwear;
- (6) FIGS. 1E-1G are enlarged environmental views showing radii associated with the first, second, and third centroids and respectively taken at Areas 1E-1G of FIG. 1D;
- (7) FIG. 2 is a bottom perspective view of an example of an article of footwear according to the present disclosure;
- (8) FIG. 3 is a bottom plan view of the article of footwear of FIG. 2;
- (9) FIG. 4 is an enlarged fragmentary view of the article of footwear of FIG. 2, showing a forefoot region of the article of footwear;
- (10) FIG. 5 is an environmental view of a forefoot plate of the article of footwear of FIG. 2 overlaid with the centroids and radii of rotation associated with the first, second, and third movements of FIG. 1D;
- (11) FIGS. 5A-5C are enlarged environmental views respectively taken at Areas 5A-5C of FIG. 5 and showing the radii of rotation associated with the first, second, and third movements;
- (12) FIG. 6 is a bottom perspective view of an example of an article of footwear according to the present disclosure;
- (13) FIG. 7 is a bottom plan view of the article of footwear of FIG. 6;
- (14) FIG. 8 is an enlarged fragmentary view of the article of footwear of FIG. 6, showing a forefoot region of the article of footwear;
- (15) FIG. 9 is a bottom perspective view of an example of an article of footwear according to the present disclosure;
- (16) FIG. 10 is a bottom plan view of the article of footwear of FIG. 9; and
- (17) FIG. 11 is an enlarged fragmentary view of the article of footwear of FIG. 9, showing a forefoot region of the article of footwear;
- (18) Corresponding reference numerals indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION

- (19) Example configurations will now be described more fully with reference to the accompanying drawings. Example configurations are provided so that this disclosure will be thorough, and will

fully convey the scope of the disclosure to those of ordinary skill in the art. Specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of configurations of the present disclosure. It will be apparent to those of ordinary skill in the art that specific details need not be employed, that example configurations may be embodied in many different forms, and that the specific details and the example configurations should not be construed to limit the scope of the disclosure.

(20) The terminology used herein is for the purpose of describing particular exemplary configurations only and is not intended to be limiting. As used herein, the singular articles “a,” “an,” and “the” may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “comprising,” “including,” and “having,” are inclusive and therefore specify the presence of features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. Additional or alternative steps may be employed.

(21) When an element or layer is referred to as being “on,” “engaged to,” “connected to,” “attached to,” or “coupled to” another element or layer, it may be directly on, engaged, connected, attached, or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly engaged to,” “directly connected to,” “directly attached to,” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.). As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(22) The terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections. These elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as “first,” “second,” and other numerical terms do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example configurations.

(23) The details of one or more implementations of the disclosure are set forth in the accompanying drawings and the description below. Other aspects, features, and advantages will be apparent from the description, the drawings, and the claims.

(24) In one aspect of the disclosure, a structure for an article of footwear includes a first plurality of traction elements having a first series of directional traction elements arranged within a first annular zone in a first rotational direction around a pivot zone. The sole structure further includes a second plurality of traction elements including a second series of directional traction elements arranged within a second annular zone concentric with and larger than the first annular zone. The second plurality of traction elements are arranged in the first rotational direction around the pivot zone.

(25) Aspects of the disclosure may include one or more of the following optional features. In some examples, each of the directional traction elements is elongate and includes a length extending from a first end to a second end along the first rotational direction. In some implementations, each of the directional traction elements includes a chamfer formed adjacent to at least one of the first end and the second end.

(26) In some configurations, each of the directional traction elements includes a concave inner surface and a convex outer surface each extending along the first rotational direction. Optionally,

the inner surface converges with the outer surface along the first rotational direction.

(27) In some implementations, the sole structure includes a third plurality of directional traction elements disposed in a heel region, each of the directional traction elements of the third plurality of directional traction elements oriented in the first rotational direction. In some examples, the first plurality of traction elements and the second plurality of traction elements are disposed in a forefoot region of the sole structure.

(28) In some configurations, the first plurality of traction elements further includes an omnidirectional traction element arranged within the first annular zone. Here, the omnidirectional traction element is disposed on a medial side of the sole structure and at least one of the directional traction elements of the first series is disposed on a lateral side of the sole structure. In some examples, at least one of directional traction elements includes a unidirectional traction element and at least one of the directional traction elements includes a bidirectional traction element.

(29) Another aspect of the disclosure provides a sole structure for an article of footwear including a first annular group of traction elements and a second annular group of traction elements. The first annular group of traction elements is arranged in series along a first annular zone in a forefoot region. The first annular group includes a first directional traction element on a lateral side of the sole structure and a second directional traction element on a medial side of the sole structure. The second annular group of traction elements is arranged in series along a second annular zone concentric with the first annular zone. The second annular group of traction elements includes a third directional traction element on the lateral side of the sole structure and a fourth directional traction element on the medial side of the sole structure.

(30) Aspects of the disclosure may include one or more of the following optional features. In some examples, each of the directional traction elements is elongate and includes a length extending from a first end to a second end along a first rotational direction around a pivot zone of the sole structure. Optionally, each of the directional traction elements includes a chamfer formed adjacent to at least one of the first end and the second end.

(31) In some implementations, each of the directional traction elements includes a concave inner surface and a convex outer surface each extending along the first rotational direction. In some examples, the inner surface converges with the outer surface along the first rotational direction. Optionally, the sole structure includes a third group of directional traction elements disposed in a heel region, each of the directional traction elements of the third group of directional traction elements oriented in the first rotational direction. In some configurations, the first annular group and the second annular group are disposed in the forefoot region of the sole structure.

(32) In some examples, the first annular group of traction elements further includes an omnidirectional traction element arranged along the first annular zone. Here, the omnidirectional traction element may be disposed on a medial side of the sole structure. In some implementations, at least one of directional traction elements includes a unidirectional traction element and at least one of the directional traction elements includes a bidirectional traction element.

(33) FIGS. 1A-1F illustrate examples of reactionary forces and motions corresponding to common athletic movements associated with an article of footwear. As shown, these forces and motions are illustrated with respect to an article of footwear including a conventional pattern of traction elements configured for translational (i.e., lateral, longitudinal) traction with a ground surface. In FIG. 1A, a first contact zone Z.sub.P1 associated with a 45° outside cut (i.e., forward and lateral direction) is shown. Here, the first contact zone Z.sub.P1 indicates a pressure area along the anterior-medial side of the article of footwear, with a higher degree of contact at the anterior end associated with the 45° movement. A centroid C.sub.P1 associated with the first contact zone Z.sub.P1 is located adjacent to an anterior end of the article of footwear. FIG. 1B illustrates a second contact zone Z.sub.P2 and second centroid C.sub.P2 corresponding to a second movement associated with a 180° outside cut (i.e., lateral direction). As shown, the second contact zone Z.sub.P2 and second centroid C.sub.P2 are shifted away from the anterior end and towards the

lateral side relative to the first contact zone Z.sub.P1 and first centroid C.sub.P1. FIG. 1C illustrates a third contact zone Z.sub.P3 and third centroid C.sub.P3 corresponding to a third movement associated with a 90° inside cut (i.e., longitudinal direction), such as an acceleration, deceleration, or planting kick. Here, the third contact zone Z.sub.P3 includes a full width of the sole structure such that the third centroid C.sub.P3 is closer to the lateral side than the first and second centroids C.sub.P1, C.sub.P2. For clarity, FIGS. 1A-1C each show a plurality of radii of rotation associated with the respective centroids C.sub.P1, C.sub.P2, C.sub.P3, which will be discussed in greater detail with respect to FIGS. 1D-1G. As discussed throughout the application, the centroids C.sub.P1, C.sub.P2, C.sub.P3 may collectively define a rotational or pivot zone Z.sub.P, which is an area of the sole structure encompassing all three of the centroids C.sub.P1, C.sub.P2, C.sub.P3 about which the sole structure may pivot during any one of the three movements.

(34) FIGS. 1D-1G illustrate the relationships between the rotational motions corresponding to each of the first, second, and third centroids C.sub.P1, C.sub.P2, C.sub.P3 associated with the first, second, and third movements. In FIG. 1D, the rotational radii corresponding to each of the centroids C.sub.P1, C.sub.P2, C.sub.P3 are all overlaid upon the same example sole structure. As shown, the rotational radii are offset relative to one another such that the rotational radii have different degrees of alignment at different areas of the sole structure. For example, at Area 1E (FIG. 1E), the rotational radii associated with the first centroid C.sub.P1 (solid line) and the third centroid C.sub.P3 (dashed line) have a relatively high degree of alignment (e.g., tangentially) to each other while the rotational radius associated with the second centroid C.sub.P2 (solid line) extends transversely to the rotational radii associated with the second centroid C.sub.P2 and the third centroid C.sub.P3. Accordingly, at Area 1E, the sole structure moves in a similar rotational direction during a 45° outside cut and a 90° inside cut, but moves along a different rotational path during a 180° outside cut. In another example, at Area 1F (FIG. 1F), the rotational radii associated with the first, second, and third centroids C.sub.P1, C.sub.P2, C.sub.P3 have a relatively low degree of alignment with each other. Thus, the sole structure moves in different rotational directions at Area 1F during a 45° outside cut, a 180° outside cut, and a 90° inside cut. At Area 1G (FIG. 1G) the rotational radii associated with the first, second, and third centroids C.sub.P1, C.sub.P2, C.sub.P3 have a relatively high degree of alignment with each other, indicating that the sole structure moves along a common rotational path at Area 1G during a 45° outside cut, a 180° outside cut, and a 90° inside cut.

(35) With continued reference to FIGS. 1D-1G, the different degrees of alignment between the rotational radii of the first, second, and third centroids C.sub.P1, C.sub.P2, C.sub.P3 may affect a torsional force associated with the article of footwear based on the shape, location, and orientation of traction elements on the article of footwear. For instance, the example of the article of footwear shown in FIGS. 1D-1G includes chevron-shaped traction elements arranged in a random pattern with broad faces of the traction elements being transverse to the rotational radii. Thus, in areas of the sole structure having a relatively high degree of alignment (e.g., Areas 1E, 1G), conventional configurations and shapes of traction elements may impart a higher torsional force during the associated movements (e.g., cuts, kicks, etc.) as the traction elements engage the ground surface along the direction of the rotational radii.

(36) As discussed in greater detail below, the examples of the articles of footwear **10**, **10a**, **10b** according to the present disclosure are configured to tune torsional forces associated with the articles of footwear **10**, **10a**, **10b** by providing an annular series of traction elements that are aligned with one another based on an optimized alignment with the rotational radii. Here, combinations of directional traction elements and omnidirectional traction elements are incorporated based on the relationship between the rotational radii. For instance, directional traction elements are provided in areas of the sole structure associated with relatively high degrees of alignment between rotational radii, while omnidirectional traction elements are provided in areas of the sole structure associated with relatively low degrees of alignment between rotational radii.

(37) Referring to FIG. 2, an article of footwear **10** includes a sole structure **100** and an upper **200** attached to the sole structure **100**. The footwear **10** may further include an anterior end **12** associated with a forward-most point of the footwear, and a posterior end **14** corresponding to a rearward-most point of the footwear **10**. As shown in FIG. 3, a longitudinal axis A.sub.10 of the footwear **10** extends along a length of the footwear **10** from the anterior end **12** to the posterior end **14** parallel to a ground surface, and generally divides the footwear **10** into a medial side **16** and a lateral side **18**. Accordingly, the medial side **16** and the lateral side **18** respectively correspond with opposite sides of the footwear **10** and extend from the anterior end **12** to the posterior end **14**. As used herein, a longitudinal direction refers to the direction extending from the anterior end **12** to the posterior end **14**, while a lateral direction refers to the direction transverse to the longitudinal direction and extending from the medial side **16** to the lateral side **18**.

(38) The article of footwear **10** may be divided into one or more regions. The regions may include a forefoot region **20**, a mid-foot region **22**, and a heel region **24**. The forefoot region **20** may be subdivided into a toe portion **20.sub.T** corresponding with phalanges and a ball portion **20.sub.B** associated with metatarsal bones of a foot. As shown, the article of footwear **10** may be described in terms of a metatarsophalangeal (MTP) axis A.sub.MTP corresponding to an MTP joint of the foot, which generally extends between the toe portion **20.sub.T** and the ball portion **20.sub.B**. The mid-foot region **22** may correspond with an arch area of the foot, and the heel region **24** may correspond with rear portions of the foot, including a calcaneus bone.

(39) With reference to FIGS. 2-4, the sole structure **100** includes a forefoot plate **102** attached to the upper **200** in the forefoot region **20** and a heel plate **104** attached to the upper **200** in the heel region **24**. The sole structure **100** and/or the plates **102**, **104** of the sole structure **100** may be described as including a top surface facing the upper and a bottom surface **108** formed on an opposite side of the sole structure **100** from the upper **200**. An outer peripheral edge **110** connects the top surface **106** to the bottom surface **108** and defines an outer peripheral profile of the sole structure **100**.

(40) In the illustrated example, the forefoot plate **102** and the heel plate **104** are formed as separate components such that the forefoot plate **102** extends from a first end **112** at the anterior end **12** to a second end **114** adjacent to the mid-foot region **22**. Here, a portion of the peripheral edge **110** defining the second end **114** of the forefoot plate **102** extends along a concave path from the medial side **16** to the lateral side **18** such that the second end **114** includes an arcuate recess **116** defining a pair of posterior-facing lobes **118** on opposite sides of the second end **112**. The heel plate extends from a first end **120** adjacent to the mid-foot region **22** to a second end **122** at the posterior end **14**. While the illustrated example includes the forefoot plate **102** and the heel plate **104** as separate components attached at opposite ends of the sole structure **100**, the plates **102**, **104** may be provided as a unitary component extending along an entire length of the article of footwear **10** from the anterior end **12** to the posterior end **14**.

(41) In the illustrated example, the sole structure **100** includes a plurality of directional traction elements **124-124f** and an omnidirectional traction element **126**. In this example, the directional traction elements **124-124f** include unidirectional traction elements **124-124f** provided as elongate members configured to move translationally through a ground surface (e.g., turf, soil) with a lower directional force in one direction than in an opposite direction, while the omnidirectional traction element **126** is provided as a round (i.e., cylindrical, conical, hemispherical) member configured to move through the ground surface in all directions with a substantially similar directional force.

(42) With reference to FIGS. 2-4, each of the forefoot plate **102** and the heel plate **104** includes a plurality of the unidirectional traction elements **124-124f**. As described herein, the unidirectional traction elements **124-124f** may be alternatively referred to as blade cleats **124-124f**. The blade cleats **124-124f** may each be described as having a height extending from a proximal end **130** at the bottom surface **108** of the sole structure **100** to a distal end **132** spaced apart from the bottom surface **108** of the sole structure **100**. Thus, the proximal end **130** of each blade cleat **124-124f**

forms a base **130** of the blade cleat **124-124f** while the distal end **132** forms a tip **132** of the blade cleat **124-124f** configured to engage the ground surface.

(43) With continued reference to FIGS. **2-4**, a length of each of the blade cleats **124-124f** extends from a first end **134** to a second end **136** disposed at an opposite end of the blade cleat **124-124f** from the first end **134**. Each blade cleat **124-124f** further includes a pair of side surfaces **138**, **140** formed on opposite sides of the blade cleat **124-124f** and extending from the first end **134** to the second end **136**. Accordingly, a width of each blade cleat **124-124f** is defined by a distance between the side surfaces **138**, **140**. As shown, a width of each of the blade cleats **124-124f** tapers along a direction from the second end **136** to the first end **134**. Here, the width of each blade cleat **124-124f** tapers continuously along the entire length of the blade cleat **124-124f** from the second end **136** to the first end **134**. In other words, a width of the each blade cleat **124-124f** is greater at the second end **136** than at the first end **134** such that the blade cleat **124-124f** is configured to move through a ground surface material (e.g., soil) in a direction from the first end **134** to the second end **136** with a lower resistance than in a direction from the second end **136** to the first end **134**.

(44) In the illustrated example, each of the first surface **138** and the second surface **140** may be multi-faceted such that the blade cleats **124-124f** each bend along a direction from the first end **134** to the second end **136**. For instance, the first side surface **138** may include a first plurality of facets **142** arranged in series from the first end **134** to the second end **136**. The facets **142** of the first side surface **138** are angled towards each other and cooperate to form a cupped or concave first surface **138**, which may be referred to as an inner surface **138** of the blade cleat **124-124f**. Conversely, the second surface **140** may include a second plurality of facets **144** arranged in series from the first end **134** to the second end **136** on the opposite side of the blade cleat **124-124f** from the first surface **138**. The facets **144** of the second surface **140** are angled away from each other and cooperate to provide the second surface **140** with a convex shape. Thus, the second surface **140** may be referred to as an outer surface **140**. In the illustrated example, the inner surface **138** and the outer surface **140** each include a pair of the facets **142**, **144** such that each blade cleat **124-124f** may be described as including first and second segments. However, in other examples, a facet resolution of the inner surface **138** and/or the outer surface **140** may be increased such that the surfaces **138**, **140** include a greater number of facets **142**, **144** or are fully arcuate.

(45) As discussed in greater detail below, the inner and outer surfaces **138**, **140** of the blade cleats **124-124f** are configured to be aligned along one or more rotational radii of the sole structure **100** such that the surfaces are substantially aligned along one or more rotational paths associated with the pivot zone Z.sub.P. Thus, the elongate shapes of the blade cleats **124-124f** provided by the tapering width and the curved surfaces **138**, **140** facilitate movement of the blade cleats **124-124f** through the ground surface along the direction of the side surfaces **138**, **140** with relatively low resistance while providing a high level of resistance (i.e., traction) in a direction transverse to the side surfaces **138**, **140**.

(46) Optionally, each of the blade cleats **124-124f** may include a chamfer **146** connecting the distal end **132** and the first end **134** of the blade cleat **124-124f**. When included, the chamfer **146** includes a surface formed at an oblique angle between the distal end **132** and the first end **134** of the blade cleat **124-124f**. The chamfer **146** provides the blade cleat **124-124f** with a shorter length at the distal end **132** of the blade cleat **124-124f** than at the base **130** of the blade cleat **124-124f** such that the blade cleat **124-124f** is configured to progressively engage the ground surface as the blade cleat **124-124f** is inserted into the ground surface.

(47) In some examples, the blade cleats **124-124f** include caps **148** attached at the distal end **132** and, when present, the chamfer **146**. Here, the caps **148** include a different material than the blade cleat **124-124f** and are configured to tune an interface between the blade cleats **124-124f** and the ground surface. For instance, the caps **148** may include materials having a lower durometer or a higher coefficient of friction than the body of the blade cleat **124-124f** to provide the blade cleats **124-124f** with better traction on relatively hard ground surfaces. Alternatively, the caps **148** may

include materials having a higher durometer than a material of the blade cleats **124-124f** to provide each of the blade cleats **124-124f** with a hard tip for engaging softer ground surfaces.

(48) With continued reference to FIGS. **2-4**, the omnidirectional traction element **126** has a height extending from a proximal end **150** at the bottom surface **108** of the sole structure **100** to a distal end **152** spaced apart from the bottom surface **108** of the sole structure **100**. Thus, the proximal end **150** forms a base **150** of the omnidirectional traction element **126** and the distal end **152** forms a tip **152** of the omnidirectional traction element.

(49) Unlike the unidirectional traction elements **124-124f**, which are substantially elongate in shape, the omnidirectional traction element **126** has a length and width that are substantially similar such that the omnidirectional traction element **126** is configured to move through the ground surface in all directions with substantially equal force or resistance. In the illustrated example, the omnidirectional traction element **126** is configured as a post cleat **126** having a substantially flat distal end **152**. Specifically, the post cleat **126** is frustoconical such that a width or diameter of the post cleat **126** tapers along a direction from the base **150** to the tip **152**. In other examples, the post cleat **126** may be cylindrical, hemispherical, or have an equilateral polygonal cross section.

(50) Referring to FIGS. **3** and **4**, the forefoot plate **102** of the present example includes a first annular cleat group **154** disposed generally in the toe portion **20.sub.T** of the forefoot region **20** and a second annular cleat group **156** arranged through the ball portion **20.sub.B** of the forefoot region **20**. Here, the first annular cleat group **154** includes a plurality of the blade cleats **124a-124d** and a single one of the post cleats **126** all arranged in series within a first annular zone **Z.sub.154** circumscribing the central pivot zone **Z.sub.P** associated with the centroids **C.sub.P1**, **C.sub.P2**, **C.sub.P3** discussed above. Similarly, the second annular cleat group **156** includes a pair of the blade cleats **124e**, **124f** arranged in series within a second annular zone **Z.sub.156** that is concentric with the first annular zone **Z.sub.154**. Here, the first annular zone **Z.sub.154** is formed between a first minor radius **R.sub.154a** and a first major radius **R.sub.154b** and the second annular zone **Z.sub.156** is formed between a second minor radius **R.sub.156a** that is greater than the first major radius **R.sub.154a** and a second major radius **R.sub.156b**. Thus, while the traction elements **124a-124f**, **126** of each of the annular cleat groups **154**, **156** are not aligned along a common radius, the traction elements **124a-124f**, **126** of each cleat group **154**, **156** are aligned within a radius range defined between the minor radii **R.sub.154a**, **R.sub.156a** and the major radii **R.sub.154b**, **R.sub.156b**.

(51) The first annular cleat group **154** may be referred to as an inner annular cleat group **154** and includes the post cleat **126** disposed immediately adjacent to the peripheral edge **110** on the medial side **16**. The post cleat **126** is disposed between the anterior end **12** and the MTP axis **A.sub.MTP**. The inner annular cleat group **154** includes four of the blade cleats **124a-124d** arranged in series around the first annular zone **Z.sub.154**. As shown, the blade cleats **124a-124d** are arranged in the same rotational direction around the first annular zone **Z.sub.154** such that the first ends **134** of each one of the blade cleats **124b-124d** face the second ends **136** of a preceding one of the blade cleats **124a-124c** while the inner surface **138** of each blade cleat **124a-124d** faces inwardly towards the pivot zone **Z.sub.P**. Thus, the blade cleats **124a-124d** are arranged to move in a rotational direction around the pivot zone **Z.sub.P**.

(52) Starting at the post cleat **126**, the blade cleats **124a-124d** are arranged in order including a first blade cleat **124a** on the medial side **16** of the anterior end **12**, a second blade cleat **124b** on the lateral side **18** of the anterior end **12**, a third blade cleat **124c** immediately adjacent to the peripheral edge **110** on the lateral side **18**, and a fourth blade cleat **124d** adjacent to the longitudinal axis **A.sub.10** and the MTP axis **A.sub.MTP**. As shown in FIG. **4**, the first blade cleat **124a** is oriented such that the first end **134** faces the post cleat **126** on the medial side **16** and the outer surface **140** of the first blade cleat **124a** is adjacent and substantially parallel to the peripheral edge **110** of the forefoot plate **102**. The first end **134** of the second blade cleat **124b** faces the second end **136** of the first blade cleat **124a** and the outer surface **140** converges with the peripheral edge **110** along a

direction from the first end **134** to the second end **136**. Thus, the first end **134** of the second blade cleat **124b** is spaced apart from the peripheral edge **110** by a greater distance at the first end **134** than at the second end **136**. The third blade cleat **124c** is disposed immediately adjacent to the peripheral edge **110** on the lateral side **18** and diverges from the peripheral edge **110** along the direction from the first end **134** to the second end **136**. The fourth blade cleat **124d** is disposed in a central portion of the sole structure **100** adjacent to the longitudinal axis A.sub.10 and is oriented such that the first end **134** is closer to the anterior end **12** and the medial side **16** than the second end **136**.

(53) With continued reference to FIGS. **3** and **4**, the second annular cleat group **156** includes a pair of the blade cleats **124e**, **124f** arranged around the second annular zone Z.sub.156. As provided above, the second annular zone Z.sub.156 is concentric with the first annular zone Z.sub.154 and has a larger radius than the first annular zone Z.sub.154 such that the blade cleats **124e**, **124f** of the second annular cleat group **156** partially surround the first annular cleat group **154**. Here, the second annular zone Z.sub.156 is sized such that the blade cleats **124e**, **124f** of the second annular cleat group **156** are disposed on a posterior side of the MTP axis A.sub.MTP (i.e., in the ball portion **20.sub.B**). The blade cleats **124e**, **124f** of the second annular cleat group **156** are arranged in the same rotational direction (e.g., clockwise, counterclockwise) around the pivot zone Z.sub.P as the blade cleats **124a-124f** of the first annular cleat group **154**. For example, the first ends **134** of each of the blade cleats **124e**, **124f** of the second annular cleat group **156** face the lateral side **18** of the sole structure **100**. A first one of the blade cleats **124e** of the second annular cleat group **156** is disposed adjacent to the medial side **16** and a second one of the blade cleats **124f** of the second annular cleat group **156** is disposed adjacent to the lateral side **18**. The inner surfaces **138** of each of the blade cleats **124e**, **124f** are oriented towards the pivot zone Z.sub.P.

(54) With reference to FIGS. **5-5C**, the positions of one of the inner blade cleats **124c**, one of the outer blade cleats **124e**, and the post cleat **126** are illustrated relative to the radii of rotation associated with the pivot centroids C.sub.P1, C.sub.P2, C.sub.P3. As provided above, the first pivot centroid C.sub.P1 (FIG. **1A**) is associated with a 45° outside cut, the second pivot centroid C.sub.P2 (FIG. **1B**) is associated with a 180° outside cut (FIG. **1B**), and the third pivot centroid C.sub.P3 (FIG. **1C**) is associated with a 90° inside cut (e.g., accelerating, decelerating, or a planting kick).

(55) With reference to FIG. **5A**, the third blade cleat **124c** of the inner annular cleat group **154** is shown with the radii of rotation associated with the pivot centroids C.sub.P1, C.sub.P2, C.sub.P3 overlaid. Here, the length of the third blade cleat **124c** (i.e., from the first end **134** to the second end **136**) is substantially tangentially aligned along the radii of rotation associated with the first pivot centroid C.sub.P1 and the third pivot centroid C.sub.P3 and is oriented at an acute angle relative to the radius of rotation associated with the second pivot centroid C.sub.P2. Thus, the third blade cleat **124c** is configured to move in the rotational directions corresponding to the first and third centroids C.sub.P1, C.sub.P3.

(56) FIG. **5B** shows the position of the post cleat **126** of the inner annular cleat group **154** with the radii of rotation of the pivot centroids C.sub.P1, C.sub.P2, C.sub.P3 overlaid. As shown, in Area **5B**, the radii of rotation of the pivot centroids C.sub.P1, C.sub.P2, C.sub.P3 are substantially misaligned such that there is no best-fit orientation for a unidirectional blade cleat **124**. In other words, incorporating a directional traction element at Area **5B** to accommodate rotational movement about one of the centroids C.sub.P1, C.sub.P2, C.sub.P3 would result in a relatively high degree of rotational resistance (i.e., traction) to rotational movement about the other two centroids C.sub.P1, C.sub.P2, C.sub.P3. Accordingly, the omnidirectional post cleat **126** is placed at Area **5B** to accommodate each of the different radii of rotation of the centroids C.sub.P1, C.sub.P2, C.sub.P3.

(57) In FIG. **5C**, the medial side blade cleat **124e** of the outer annual cleat group **156** is shown with the radii of rotation of the pivot centroids C.sub.P1, C.sub.P2, C.sub.P3 overlaid. Here, the radii of

rotation of the pivot centroids C.sub.P1, C.sub.P2, C.sub.P3 have a relatively high degree of tangential alignment such that the medial blade cleat **124e** is oriented to align the inner and outer surfaces **138**, **140** with each of the radii of rotation of the pivot centroids C.sub.P1, C.sub.P2, C.sub.P3. Thus, the medial blade cleat **124e** is configured to move in the rotational directions corresponding to the first, second, and third centroids C.sub.P1, C.sub.P2, C.sub.P3.

(58) With continued reference to FIG. 5, the blade cleats **124e-124f** of the forefoot plate **102** are positioned at areas of the forefoot plate **102** where the radii of rotation of at least two of the pivot centroids C.sub.P1, C.sub.P2, C.sub.P3 are aligned. For instance, in addition to the blade cleats **124c**, **124e** discussed with respect to FIGS. 5A and 5C, FIG. 5 shows the first blade cleat **124a** of the inner annular cleat group **154** located where the radii of rotation for the first and third pivot centroids C.sub.P1, C.sub.P3 are aligned while the second blade cleat **124b** of the inner annular cleat group **154** is located where the radii of rotation for the second and third pivot centroids C.sub.P1, C.sub.P2 are aligned. The fourth blade cleat **124d** of the inner annular cleat group **154** and the lateral blade cleat **124f** of the outer annular cleat group **156** are each positioned at areas with relatively high degrees of alignment between the radii of rotation of all three of the pivot centroids C.sub.P1, C.sub.P2, C.sub.P3.

(59) As shown in FIG. 3, the heel plate **104** includes a plurality of the blade cleats **124** each oriented along the same rotational direction about the pivot zone Z.sub.P as the blade cleats **124a-124f** of the forefoot plate **102**. In the illustrated example, the heel plate **104** includes two pairs of the blade cleats **124**, with a first pair aligned longitudinally along the lateral side **18** of the longitudinal axis A.sub.10 and a second pair aligned longitudinally along the medial side **16** of the longitudinal axis A.sub.10. Here, all of the blade cleats **124** of the heel plate **104** include the first ends **134** facing towards the lateral side **18** of the sole structure **100**. The blade cleats **124** on the medial side **16** of the longitudinal axis A.sub.10 are oriented such that the second end **136** faces the medial side **16** and is closer to the anterior end **12** than the first end **134**. Similarly, the blade cleats **124** on the lateral side **18** of the longitudinal axis A.sub.10 are oriented such that the first end **134** faces the lateral side **18** and is closer to the anterior end **12** than the second end **136**.

(60) As provided above, all of the blade cleats **124-124f** of the forefoot plate **102** and the heel plate **104** are oriented in the same rotational direction (i.e., clockwise, counterclockwise) such that the inner surfaces **138** and the outer surfaces **140** are substantially tangential to concentric radii of rotation about the pivot zone Z.sub.P. Thus, the tapered, elongate, and bent shapes of the blade cleats **124** allow the sole structure **100** to rotate about the pivot zone Z.sub.P with a minimized torsional force while the inner surfaces **138** and the outer surfaces **140** of the blade cleats **124** are configured to provide increased traction in lateral and longitudinal directions of the sole structure **100**.

(61) With particular reference to FIGS. 6-8, an article of footwear **10a** is provided and includes a sole structure **100a** and the upper **200** attached to the sole structure **100a**. In view of the substantial similarity in structure and function of the components associated with the article of footwear **10** with respect to the article of footwear **10a**, like reference numerals are used hereinafter and in the drawings to identify like components while like reference numerals containing letter extensions are used to identify those components that have been modified.

(62) In the example of the sole structure **100a** shown in FIGS. 6-8, a forefoot plate **102a** includes an inner annular cleat group **154a** and an outer annular cleat group **156a** that are similar to the inner annular cleat group **154** and the outer annular cleat group **156** discussed previously. However, in this example, the radii R.sub.154a, R.sub.154b, R.sub.156a, R.sub.156b of the annular zones Z.sub.154, Z.sub.156 have been reduced such that the inner and outer annular cleat groups **154a**, **156a** are more tightly grouped around the pivot zone Z.sub.P. For example, the post cleat **126** and the first, second, and third blade cleats **124a-124c** are offset inwardly from the peripheral edge **110** of the sole structure **100**. Furthermore, the blade cleats **124e**, **124f** of the outer annular cleat group **156a** are each positioned on an anterior side of the MTP axis A.sub.MTP (i.e., in the toe portion

20r).

(63) By reducing the respective radii of the cleat groups **154a**, **154b**, a torsional force required to rotate or twist the cleat groups **154a**, **154b** about the pivot zone Z.sub.P through the ground surface is reduced relative to a torsional force associated with the sole structure **100** discussed previously. However, while the torsional forces associated with the sole structure **100a** are reduced, the rotational configuration of the traction elements **124**, **126** provides translational (e.g., lateral, longitudinal) traction forces comparable to the forces provided by the sole structure **100**, as the inner and outer surfaces **138**, **140** of the blade cleats **124** cooperate to engage the ground surface in different translational directions.

(64) With particular reference to FIGS. **9-11**, an article of footwear **10b** is provided and includes a sole structure **100b** and the upper **200** attached to the sole structure **100b**. In view of the substantial similarity in structure and function of the components associated with the article of footwear **10** with respect to the article of footwear **10b**, like reference numerals are used hereinafter and in the drawings to identify like components while like reference numerals containing letter extensions are used to identify those components that have been modified.

(65) The sole structure **100b** of FIGS. **9-11** is substantially similar to the sole structure **100** discussed above and includes an inner annular cleat group **154b** and an outer annular cleat group **156b** arranged around annular zones Z.sub.154b, Z.sub.156b. As with the sole structure **100** discussed above, the inner annular cleat group **154b** is arranged along an annular zone Z.sub.154 such that the traction elements **124b-124d**, **160a-160b** of the inner annular cleat group **154b** are adjacent to the peripheral edge **110** of the forefoot plate **102b** and the traction elements **124f**, **160c** of the outer annular cleat group **156b** are positioned on a posterior side of the MTP axis A.sub.MTP (i.e., in the ball portion 20.sub.B).

(66) The sole structure **100b** of the present example includes a plurality of bi-directional traction elements **160a-160c** in addition to the unidirectional traction elements **124b-124d**, **124f**. For example, where the sole structure **100** discussed above includes the post cleat **126**, the first blade cleat **124a** of inner annular cleat group **154**, and the medial blade cleat **124e** of the outer annular cleat group **156**, the sole structure **100b** of the present example includes bi-directional traction elements **160a-160c** incorporated into the annular cleat groups **154b**, **156b**. Generally, the bi-directional traction elements **160a-160c** are formed as symmetrical wing cleats **160a-160c** configured to allow bi-directional movement along a lengthwise direction of the traction element **160a-160c**. The bi-directional traction elements **160a-160c** and the unidirectional traction elements **124b-124d**, **124f** may be collectively referred to as directional traction elements.

(67) With reference to FIG. **9**, each of the wing cleats **160a-160c** includes height extending from a proximal end **162** at the bottom surface **108** of the sole structure **100** to a distal end **164** spaced apart from the bottom surface **108** of the sole structure **100**. Thus, the proximal end **162** forms a base of the wing cleat **160a-160c** and the distal end **164** forms a tip of the wing cleat **160a-160c**.

(68) With continued reference to FIGS. **9-11**, a length of each of the wing cleats **160a-160c** extends from a first end **166** to a second end **168** disposed at an opposite end of the wing cleat **160a-160c** from the first end **166**. Each wing cleat **160a-160c** further includes a pair of side surfaces **170**, **172** formed on opposite sides of the wing cleat **160a-160c** and extending from the first end **166** to the second end **168**. Accordingly, a width of each wing cleat **160a-160c** is defined by a distance between the side surfaces **170**, **172**. As shown, the width of each of the wing cleats **160a-160c** tapers along the length from an intermediate portion towards each end **166**, **168**. In other words, a width of the each wing cleat **160a-160c** is greater in a central portion than at the first end **166** and second end **168** such that the wing cleat **160** is configured to slice through a ground surface material (e.g., soil) in a direction towards either end **166**, **168**.

(69) In the illustrated example, each of the first side surface **170** and the second side surface **172** may be multi-faceted such that the wing cleats **160a-160c** each bend along a direction from the first end **166** to the second end **168**. For instance, the first side surface **170** may include a first plurality

of facets **174** arranged in series from the first end **166** to the second end **168**. The facets **174** of the first side surface **170** are angled towards each other and cooperate to form a cupped or concave first side surface **170**, which may be referred to as an inner surface **170** of the wing cleat **160a-160c**. Conversely, the second surface **172** may include a second plurality of facets **176** arranged in series from the first end **166** to the second end **168** on the opposite side of the wing cleat **160a-160c** from the first side surface **170**. The facets **176** of the second side surface **172** are angled away from each other and cooperate to provide the second side surface **172** with a convex shape. Thus, the second side surface **172** may be referred to as an outer surface **172**. In the illustrated example, the inner side surface **170** and the outer side surface **172** each include three of the facets **174**, **176** such that each wing cleat **160a-160c** may be described as including first and second end segments extending from an intermediate segment. However, in other examples, a facet resolution of the inner surface **170** and/or the outer surface **172** may be increased such that the surfaces **170**, **172** include a greater number of facets **174**, **176** or are fully arcuate.

(70) Optionally, each of the wing cleats **160a-160c** may include a pair of chamfers **178** connecting the distal end **164** and with each of the first end **166** and the second end **168**. When included, the chamfers **178** include a surface formed at an oblique angle relative to the distal end **164** and each of the first end **166** and the second end **168** of the wing cleat **160a-160c**. The chamfers **178** provide the wing cleat **160a-160c** with a shorter length at the distal end **164** of the wing cleat **160a-160c** than at the base **162** of the wing cleat **160** such that the wing cleat **160** is configured to progressively engage the ground surface as the wing cleat **160a-160c** is inserted into the ground surface.

(71) In some examples, the wing cleats **160a-160c** include caps **180** attached at the distal end **164** and, when present, the chamfers **178**. Here, the caps **180** include a different material than the body of the wing cleats **160a-160c** and are configured to tune an interface between the wing cleats **160a-160c** and the ground surface. For instance, the caps **180** may include materials having a lower durometer or a higher coefficient of friction to provide the wing cleats **160a-160c** with better traction on relatively hard ground surfaces. Alternatively, the caps **180** may include materials having a higher durometer than a material of the wing cleats **160a-160c** to provide each of the wing cleats **160a-160c** with a hard tip for engaging softer ground surfaces.

(72) The following Clauses provide an exemplary configuration of a sole structure and an article of footwear described above.

(73) Clause 1: A sole structure for an article of footwear, the sole structure comprising: a first plurality of traction elements including a first series of directional traction elements arranged within a first annular zone in a first rotational direction around a pivot zone; and a second plurality of traction elements including a second series of directional traction elements arranged within a second annular zone concentric with and larger than the first annular zone, the second plurality of traction elements arranged in the first rotational direction around the pivot zone.

(74) Clause 2: The sole structure of Clause 1, wherein each of the directional traction elements is elongate and includes a length extending from a first end to a second end along the first rotational direction.

(75) Clause 3: The sole structure of Clause 2, wherein each of the directional traction elements includes a chamfer formed adjacent to at least one of the first end and the second end.

(76) Clause 4: The sole structure of Clause 2 or 3, wherein each of the directional traction elements includes a concave inner surface and a convex outer surface each extending along the first rotational direction.

(77) Clause 5: The sole structure of Clause 4, wherein the inner surface converges with the outer surface along the first rotational direction.

(78) Clause 6: The sole structure of any one of Clauses 1-5, further comprising a third plurality of directional traction elements disposed in a heel region, each of the directional traction elements of the third plurality of directional traction elements oriented in the first rotational direction.

(79) Clause 7: The sole structure of any one of Claims 1-6, wherein the first plurality of traction elements and the second plurality of traction elements are disposed in a forefoot region of the sole structure.

(80) Clause 8: The sole structure of any one of Claims 1-7, wherein the first plurality of traction elements further includes an omnidirectional traction element arranged within the first annular zone.

(81) Clause 9: The sole structure of Claim 8, wherein the omnidirectional traction element is disposed on a medial side of the sole structure and at least one of the directional traction elements of the first series is disposed on a lateral side of the sole structure.

(82) Clause 10: The sole structure of any one of Claims 1-9, wherein at least one of directional traction elements includes a unidirectional traction element and at least one of the directional traction elements includes a bidirectional traction element.

(83) Clause 11: A sole structure for an article of footwear, the sole structure comprising: a first annular group of traction elements arranged in series along a first annular zone in a forefoot region, the first annular group including a first directional traction element on a lateral side of the sole structure and a second directional traction element on a medial side of the sole structure; and a second annular group of traction elements arranged in series along a second annular zone concentric with the first annular zone, the second annular group of traction elements including a third directional traction element on the lateral side of the sole structure and a fourth directional traction element on the medial side of the sole structure.

(84) Clause 12: The sole structure of Claim 11, wherein each of the directional traction elements is elongate and includes a length extending from a first end to a second end along a first rotational direction around a pivot zone of the sole structure.

(85) Clause 13: The sole structure of Claim 12, wherein each of the directional traction elements includes a chamfer formed adjacent to at least one of the first end and the second end.

(86) Clause 14: The sole structure of Claim 12 or 13, wherein each of the directional traction elements includes a concave inner surface and a convex outer surface each extending along the first rotational direction.

(87) Clause 15: The sole structure of Claim 14, wherein the inner surface converges with the outer surface along the first rotational direction.

(88) Clause 16: The sole structure of any one of Claims 11-15, further comprising a third group of directional traction elements disposed in a heel region, each of the directional traction elements of the third group of directional traction elements oriented in the first rotational direction.

(89) Clause 17: The sole structure of any one of Claims 11-16, wherein the first annular group and the second annular group are disposed in a forefoot region of the sole structure.

(90) Clause 18: The sole structure of any one of Claims 11-17, wherein the first annular group of traction elements further includes an omnidirectional traction element arranged along the first annular zone.

(91) Clause 19: The sole structure of Claim 18, wherein the omnidirectional traction element is disposed on a medial side of the sole structure.

(92) Clause 20: The sole structure of any one of Claims 11-19, wherein at least one of directional traction elements includes a unidirectional traction element and at least one of the directional traction elements includes a bidirectional traction element.

(93) The foregoing description has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular configuration are generally not limited to that particular configuration, but, where applicable, are interchangeable and can be used in a selected configuration, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

Claims

1. A sole structure for an article of footwear, the sole structure comprising: a first plurality of traction elements including a first series of directional traction elements including a first plurality of blade cleats arranged within a first annular zone and each of the first plurality of traction elements being oriented in a first rotational direction around a pivot zone, each blade cleat of the first plurality of blade cleats tapering continuously from a respective first end to a respective second end; and a wing cleat aligned with traction elements of the first plurality of traction elements in the first rotational direction, the wing cleat tapering from an intermediate portion to a first end and from the intermediate portion to a second end located on an opposite side of the wing cleat than the first end.
2. The sole structure of claim 1, wherein the wing cleat is disposed proximate to a medial side of the sole structure.
3. The sole structure of claim 1, wherein each of the blade cleats is elongate and includes a length extending from the respective first end to the respective second end along the first rotational direction.
4. The sole structure of claim 1, wherein each of the directional traction elements includes a chamfer formed adjacent to at least one of the respective first end and the respective second end.
5. The sole structure of claim 1, wherein each of the directional traction elements includes a concave inner surface and a convex outer surface each extending along the first rotational direction.
6. The sole structure of claim 5, wherein the inner surface converges with the outer surface along the first rotational direction.
7. The sole structure of claim 1, wherein the first plurality of traction elements and the wing cleat are disposed in a forefoot region of the sole structure.
8. The sole structure of claim 1, wherein the wing cleat is disposed between a first traction element of the first plurality of traction elements and a second traction element of the first plurality of traction elements along a length of a longitudinal axis of the sole structure.
9. The sole structure of claim 8, wherein a distance between the wing cleat and the first traction element is different than a distance between the wing cleat and the second traction element.
10. The sole structure of claim 1, wherein the wing cleat is symmetric.
11. A sole structure for an article of footwear, the sole structure comprising: a first plurality of traction elements including a first series of directional traction elements including a first plurality of blade cleats arranged within a first annular zone and each of the first plurality of traction elements being oriented in a first rotational direction around a pivot zone; and a wing cleat aligned with traction elements of the first plurality of traction elements in the first rotational direction and including a first facet, a second facet, and a third facet, the third facet disposed between the first facet and the second facet and cooperating with the first facet and the second facet to provide the wing cleat with a concave surface opposing at least one blade cleat of the first plurality of blade cleats.
12. The sole structure of claim 11, wherein the wing cleat is disposed proximate to a medial side of the sole structure.
13. The sole structure of claim 11, wherein each of the directional traction elements is elongate and includes a length extending from a first end to a second end along the first rotational direction.
14. The sole structure of claim 13, wherein each of the directional traction elements includes a chamfer formed adjacent to at least one of the first end and the second end.
15. The sole structure of claim 13, wherein each of the directional traction elements includes a concave inner surface and a convex outer surface each extending along the first rotational direction.
16. The sole structure of claim 15, wherein the inner surface converges with the outer surface along the first rotational direction.

17. The sole structure of claim 11, wherein the first plurality of traction elements and the wing cleat are disposed in a forefoot region of the sole structure.
 18. The sole structure of claim 11, wherein the wing cleat is disposed between a first traction element of the first plurality of traction elements and a second traction element of the first plurality of traction elements along a length of a longitudinal axis of the sole structure.
 19. The sole structure of claim 18, wherein a distance between the wing cleat and the first traction element is different than a distance between the wing cleat and the second traction element.
 20. The sole structure of claim 11, wherein the wing cleat is symmetric.
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